

Attach the CPU fan

This cannot be stressed enough. Ensure that the CPU fan is plugged into the correct point on the motherboard. Failing to do so could result in your CPU and motherboard being permanently damaged and all this could happen within seconds of powering up the system, leaving you with absolutely no time to realise your mistake!

Listen for diagnostic beeps

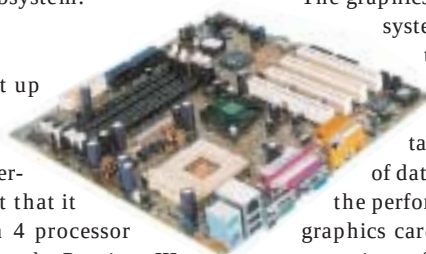
Remember to connect the motherboard speaker. Study the motherboard manual for deciphering the sequence of the diagnostic beeps that are emitted when the motherboard is powered up—a single short beep means things are in order. Otherwise, you need to diagnose what's wrong by listening to the beep sequence. Check if all fans are spinning, and that the cards and RAM modules are properly seated.

The ABC of Motherboards

The overall performance of a computer system is dependent on its ability to perform a collection of tasks optimally. It depends on various specialized subsystems to perform these different tasks—the processor subsystem, the disk subsystem, the memory subsystem and the graphics subsystem.

The processor subsystem

A motherboard is designed from chipset up to support a specific type or class of processor. The chipset it incorporates will therefore determine how the motherboard performs and the basic feature set that it offers. For instance, since the Pentium 4 processor works differently when compared to even the Pentium III, the motherboard design is varied to take advantage of the higher performance available. It needs to take advantage of the higher clock speeds at which the processors runs and also the higher (quad-pumped to 400 MHz) FSB frequency (Athlon XP requires 266 MHz of FSB). The ability to integrate a particular type of processor without compatibility issues and then to fully utilize its potential performance is the motherboards first and foremost goal.



Memory subsystem

The memory subsystem is probably the most critical subsystem because it needs to constantly feed the processor with data. To squeeze every last ounce of performance, the memory subsystem must have very tight memory timings and implementation. Despite the presence of strict memory standards, there will be slight variation in the memory modules produced by different manufacturers. The motherboard has to ensure that the memory—slots can handle these possible variations and at the same time maintain accurate and tight timing cycles to ensure error-free and high-speed data throughput.

Disk subsystem

Performance within this subsystem depends heavily on the chipsets implementation of an IDE controller. An under-performing IDE subsystem can cripple performance when it comes to video editing and other data-hungry, streaming media-based applications. Each time that these applications read or write data from or to the disk, instructions are passed to the drive's controller's device driver. The driver then pumps data through the PCI bus to the IDE controller. Improper implementation of this part of the entire subsystem can severely hamper the entire data transfer process. This happens when the PCI bus has not been isolated from noise from other devices on the motherboard which causes data corruption and lock ups due to large file transfers (this

happens when a large amount of data is suddenly pushed through the bus, which then collapses due the load).

Graphics subsystem

The graphics subsystem deals with the video performance of the system. This is the part that actually generates the images that you see on your display. The realism of computer games depends to a large extent on the total amount of bandwidth offered by AGP. More importantly, however, the graphics card needs continuous flow of data from both the processor and the memory. Although the performance of this subsystem is largely dependent on the graphics card installed, you will be amazed to find large differences in performance due to the use of different chipsets and the manner in which they are implemented in a motherboard.

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